

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- PROP-01
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(1) Research Program/Project/Study Title: Development of Calabash Cider Vinegar	
Proponent Name:	MAY JOY L. FERNANDEZ
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	Kirsty G. Tanque, Yhanie A. Lantoria, Annalie G. Campos, Anabelle F. Nobleta
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation		Sigma
Municipality/City:		Sigma
Province:		Capiz
Region:		VI
(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization Food Research

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Most of the Calabash fruits were always left rot on the ground because of its gummy smell .Yet, there are so many Calabash fruits that can be found spreading in Capiz. However, most of them were not utilized since a lot of people didn't know how to make good use of this fruit aside from making it as a juice. Add on to that, there were no sufficient souvenirs that can be found in the municipality of Sigma. With these, the researchers wanted to offer a product that would be beneficial to everyone and at the same time its very cost efficient and effective, so the study of Calabash Cider Vinegar was pursued.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study sought to answer the following:

- 1.What is the level of acceptability of Calabash Cider Vinegar in general?
2. What the level of acceptability of Calabash Cider Vinegar in terms of taste, appearance, texture and aroma?
3. What is the level of acceptability of Calabash Cider Vinegar when the respondents are classified according to sex, age and type of industry they are working?

(11) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

Research Instrument

The researcher will conduct one-on-one sensory evaluation using a survey questionnaire checklist.

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(12) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology / information from which the project proposal will take-off).

REVIEW OF RELATED LITERATURE

Juber (2022) cited that there are as many kinds of fruit vinegar as there are types of fruit. These vinegars have found important roles in recipes around the world. Fruit vinegar is tart, a little sweet, and reminiscent of the fruit that was used to make it. It also has many health benefits on top of its potent flavor.

To make it you simply let some vinegar ferment into alcohol, and then ferment again into acetic acid. This results in a sour liquid that can be used to pickle foods or add a unique flavor to recipes. Fruit vinegar is made the same way, but pieces of fruit are added at the beginning of the fermenting process. Many have declared it a cure-all in recent years, saying it can help with issues such as: infections, high blood sugar, weight loss and cancer. While many studies are still being done, here's what's currently known about how consuming fruit vinegar affects health. (Juber, 2022)

Prior Art Search

CN104087500A

Method for preparing miracle fruit vinegar

Technical field

The invention belongs to food brewing technology, in particular to a preparation method of mysterious fruit vinegar.

Background technique

Mysterious fruit, native to Brazil, has recently been introduced in China, and the juice is very sweet. However, it is easily fermented and deteriorated due to the enzyme-containing protein, so it is difficult to preserve. Therefore, how to use it quickly and maintain quality is particularly important and urgently needed to be solved.

Summary of the invention

The invention aims at the above defects, and provides a preparation method of mysterious fruit vinegar, which achieves the purpose of rapid processing and utilization of fresh

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mysterious fruit.

The object of the invention is achieved in this way.

A method for preparing a mysterious fruit vinegar, comprising the following preparation steps:

(1) Wash 10kg of fresh mysterious fruit picked with water, rinse it again with sterile water, drain the water, puncture the peel with a 1.5mm diameter steel needle, and tie each mysterious peel to 3-4 holes;

(2) Take 20kg of edible white vinegar which is fermented by liquid liquor but not sterilized, added to step (1), added 5% of acid-resistant *Saccharomyces cerevisiae* starter, and then brewed into sterile air for 12 days, 200 mesh. The pomace is filtered off, sterilized by membrane filtration, and aseptically filled to obtain a mysterious fruit vinegar.

The beneficial effects of the present invention are as follows:

(1) The freshly picked mysterious fruit is quickly cleaned, reducing the microbial contamination, and puncturing the skin, allowing the white vinegar to quickly penetrate into the pulp, causing the fruit to sink and accelerate the fermentation;

(2) The liquid fermented vinegar method using rice wine as raw material is a modern fast brewing process with short time and high efficiency. Since the raw material itself is rice wine, the white vinegar brewed is colorless and pure in taste, since it has been fermented into vinegar, After mixing with mysterious fruit, its acetic acid can quickly kill the bacteria that may be brought in by the fresh fruit, ensuring the purity, and then using vinegar has not been sterilized, containing acetic acid bacteria, and adding acid-resistant *Saccharomyces cerevisiae* starter, the mysterious fruit will be quickly Participate in the fermentation into vinegar, filter the pomace and filter it with bacteria. Without heating, the vinegar product has pure flavor, simple process, short time, good effect, high efficiency, good product quality and good solution. Mysterious fruit is easy to taste and deteriorate, and maintains the activity of functional substances such as mysterious fructose proteins and enzymes.

Detailed ways

The present invention will be further described in detail below in conjunction with specific embodiments.

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Example.

A method for preparing a mysterious fruit vinegar, comprising the following preparation steps:

(1) Wash 10kg of fresh mysterious fruit picked with water, rinse it again with sterile water, drain the water, puncture the peel with a 1.5mm diameter steel needle, and tie each mysterious peel to 3-4 holes;

(2) Take 20kg of edible white vinegar which is fermented by liquid liquor but not sterilized, added to the obtained step (1), added 5% of acid-resistant *Saccharomyces cerevisiae* starter, and then brewed into sterile air for 12 days, 200 The fruit residue is filtered, and the membrane is sterilized by filtration and aseptically filled to obtain a mysterious fruit vinegar.

The prepared product has a pure flavor and a special sweet taste of mysterious fruit, which makes the vinegar taste softer.

The above-mentioned embodiments are only illustrative of the embodiments of the present invention, and are not to be construed as limiting the scope of the invention.

Related Studies

In the study of Hadi, et.al, (2021) The effect of apple cider vinegar on lipid profiles and glycemic parameters: a systematic review and meta-analysis of randomized clinical trials, study revealed that Apple cider vinegar (ACV) is one of the three most common types of vinegar, produced by fermenting apples. This acidic solution is consumed throughout the world as a flavoring and preservative agent in foods. ACV contains a variety of flavonoids, such as gallic acid, catechin, caffeic acid, and ferulic acid. Animal experiments have reported that ACV has a variety of pharmacological functions, including anti-oxidant, anti-inflammatory, anti-diabetic, anti-hypertensive, and anti-hyperlipidemic properties. The effects of ACV on serum lipid parameters and glycemic markers have been investigated in several randomized clinical trials]; however, the results are equivocal. Indeed, some investigations have reported beneficial effects following ACV consumption on the aforementioned parameters, although others failed to detect any effects. It is conceivable that such contradictory findings might be due to the differences in study design and/or characteristics of participants (age, sex, clinical condition).

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Treatment and Procedures in making Calabash Cider Vinegar

Treatment A: 250ml Calabash Juice, 1 tablespoon bakery yeast, 1 tablespoon muscovado sugar

Treatment B. 300ml Calabash Juice, 1 tablespoon bakery yeast, 1 tablespoon muscovado sugar

Treatment C. 350ml Calabash Juice, 1 tablespoon bakery yeast, 1 tablespoon muscovado sugar

Research Design

The researchers use a experimental research.

Respondents of the Study

The respondents of the study were the people of Capiz who were working in different industries.

Data Gathering Procedure

The questionnaire will personally distribute to the respondents. The respondents fill up the information in the demographic profile honestly, then check appropriately the column in the Part II and Part III of the survey questionnaire checklist.

Data Analysis Procedure

All the data gather were tally, computed, analyze and interpret. The findings and the recommendations base on the conclusions.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

Research Objectives	Year 1						Year 2					
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Field work/observation												
2. Analysis of data gathered from observation												
3. Lay-outing, Editing and Printing of 100 copies of coffee table book												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Calabash Cider Vinegar

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(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The residents of municipality of Sigma and this product of research will be used as one of the souvenirs.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
MAY JOY L. FERNANDEZ	Study Leader	Social Science

Percent time devoted to the Program/ Project/ Study 30%

Program/Project Leader (in hours): 3 hrs

Program/Project Members (in hours): 1hr

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		2,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		3,000.00
- Meal allowance during data gathering		3,000.00
- Professional fees (laboratory analysis and other related fees)		5,000.00
- Miscellaneous (for incidental expenses)		2,000.00
- Printing		5,000.00
C. CAPITAL OUTLAY		
Total		20,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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Juber, M. (2021). Fruit Vinegar: Are There Health Benefits

(20) Capsule Curriculum Vitae (One-page brief C.V. only)